Celtic god of thunder, the Taranis was conceived as part of the UK's Future Combat Air System (FCAS) initiative to explore the technologies necessary for autonomous, stealth, and networked aerial warfare. Although it remains a technology demonstrator, the Taranis program provided the foundation for Europe's future sixth-generation combat air systems, including the Anglo-French FCAS and Tempest programs.

The Taranis' airframe design features a flying-wing configuration optimized for stealth, with sharp angular contours, radar-absorbent materials (RAM), and internal weapons bays. These elements contribute to a minimal radar cross-section (RCS) and thermal signature, enabling deep penetration in contested environments. The aircraft measures approximately 11.35 meters in length, with a 9.94-meter wingspan, and weighs about 8 tons. Powered by a single Rolls-Royce Adour Mk 951 turbofan engine, it can operate at subsonic speeds while maintaining stealth and endurance (Johnston, 2013).

Technologically, the Taranis demonstrated several groundbreaking advancements, including autonomous flight control, AI-assisted decision-making, and secure communication links for remote or semi-autonomous operations. It was capable of autonomous taxiing, takeoff, navigation, and landing, marking a major step toward fully autonomous combat aircraft. Its mission systems integrated sensor fusion, electronic warfare (EW) capabilities, and adaptive flight control algorithms, enabling real-time threat detection and mission planning without direct pilot intervention.

The internal weapon bays were designed to carry precision-guided munitions, ensuring stealth integrity while allowing for strike capability. Additionally, its AI-driven system could evaluate mission parameters, assess threats, and execute pre-programed responses autonomously – a capability that foreshadowed the operational logic of future AI-enabled UCAVs.